

Course Code: US02CBCA01

Course Title : Advanced C Programming and Introduction to Data Structures

Total Marks: 70

## Q1. Multiple Choice Questions.(Attempt All)

[10]

1. Which of the following defines a pointer variable to a character ?  
A. char &p; C. char \*p;  
B. char p; D. char p[20];
2. Which of the following is not true for pointers?  
A. Pointer stores address of another variable. C. Pointer reduces complexity of the program.  
B. Pointers are not used for Dynamic Memory Allocation. D. Pointers can be used to return multiple values from a function via function arguments
3. Given a pointer ptr to a structure named stud containing a field called sname which of the following statements correctly refer sname, if s is a structure type variable.  
A. ptr->s.sname C. ptr=\*s.sname  
B. ptr=s.sname D. None of these
4. \_\_\_\_\_ function is used to open the file.  
A. fopen() C. fileopen()  
B. open() D. file()
5. Which of the following can be used to create a new type that can be used anywhere a type is permitted?  
A. struct C. typedef  
B. array D. define
6. Which of the following keyword is used to declare a file pointer named fp.  
A. file \$fp; C. FILE \*fp;  
B. file &fp; D. FILE fp;
7. A data structure where elements can be added or removed at either end but not in the middle is known as \_\_\_\_\_.  
A. Stack C. Queue  
B. Array D. Dequeue
8. An operation that is used to delete an element from a stack is known as \_\_\_\_\_.  
A. Push C. Pop  
B. Peep D. Change
9. A storage representation of a linked list in a memory is \_\_\_\_\_.  
A. Linear C. Both A and B  
B. Non-Linear D. None of these
10. Which of the following statement is FALSE for the Queue data structure?  
A. Its nature is FIFO C. Its nature is LIFO  
B. Element is deleted from FRONT D. Element is inserted at REAR

**Q2. Fill in the Blanks and True/False**

[08]

1. \_\_\_\_\_ function is used to release the allocated memory using dynamic memory allocation functions.
2. \_\_\_\_\_ data structure allows a portion of memory to be shared by different types of data.
3. An operation that is used to insert an element from a stack is known as \_\_\_\_\_.
4. A data structure that contains not only a data field but also contains pointer field is known as \_\_\_\_\_.
5. Pointer stores address of another variable. (True/False)
6. struct keyword is used to declare a structure. (True/False)
7. Element is deleted from BOTTOM of stack. (True/False)
8. A linked list in which last node pointing to the first node is known as Circular Linked List. (True/False)

**Q3. Short Answer Questions (Attempt any TEN [2 marks each])**

[20]

1. Differentiate between "\*" and "&" Operator
2. Define pointer and give one example.
3. List different pointer declaration style. Which one is preferable?
4. Differentiate between : structure and union
5. Explain the fclose() function with example.
6. What do you mean by a file pointer?
7. Define Data Structure.
8. Differentiate between : primitive and non primitive data structure
9. What do you mean Linear Data Structure?
10. What is a doubly Linked List?
11. List types of Linked List.
12. State various types of queue.

**Q4. Long Questions (Attempt any four)**

[32]

1. Explain pointer declaration and initialization in detail with example.
2. Explain pointer to array in detail.
3. Explain structure declaration with example.
4. Explain concept of nested structure in detail.
5. Explain a STACK with an example. Write algorithm for PUSH and POP operations of STACK.
6. Write down advantages of data structure and Explain operations that can be performed on data structures in details.
7. Explain Linked list in brief and write an algorithm to insert an element at the beginning of a Singly linked list.
8. Explain Queue in brief and Write an algorithm to insert an element into a simple queue.